TheNightOps

Autonomous SRE Agent for Kubernetes

Built with Google Agent Development Kit (ADK) & Gemini

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About Me

Mehul Patel

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Google Developer Expert in Google Cloud

Docker Captain

Running Grafana & Friends Meetup in Berlin

Passionate about SRE, Kubernetes, AI-powered operations & Open Source

Agenda

The Problem	Why traditional alerting falls short
What is ADK?	Google Agent Development Kit overview
Architecture	Multi-agent design & MCP integration
Deep Dive	Agents, MCP servers, orchestration patterns
Live Demo	Triggering failures & watching agents investigate
Takeaways	What we learned building with ADK

Me at 3 AM:

"I should automate this"

Also me at 3 AM:

manually SSHing into 14 pods

The Problem

Why traditional incident response doesn't scale

The 3 AM Problem

70%	45min	60%
Alert fatigue rate in SRE teams	Average time to first meaningful action	Incidents repeat with same root cause

Alert → Triage → Investigate → Escalate → Fix → Repeat

What if an AI agent could handle the investigation autonomously?

SRE Job Description:

"Automate everything"

SRE Reality:

"kubectl get pods -A | grep -v Running"

...at 3 AM, for the 47th time this quarter

What is Google ADK?

Agent Development Kit — building blocks for multi-agent systems

Google ADK at a Glance

Open-source framework for building multi-agent AI systems

Open Source
Apache 2.0

Python, Go, TS & Java
SDK support

Model-Agnostic
Gemini, Claude, GPT-4o, Mistral

Production-Ready
Built-in eval & debugging

Multi-Agent Orchestration

Parallel, sequential & hierarchical agent orchestration with structured handoffs

Native MCP Support

Connect official & custom MCP servers for tool integration

Streaming & Real-time

Bidirectional audio/video streaming with real-time agent interaction

Local Dev & Deploy

CLI + Web UI for development, deploy to Cloud Run or GKE

Deep GCP Integration

ADK fits naturally into the Google Cloud ecosystem

Compute

Cloud Run, GKE
Compute Engine

Serverless & container deployment

Data

BigQuery, AlloyDB
Cloud Spanner, Postgres

Storage & analytics integration

Operations

Cloud Monitoring
Cloud Logging, Security

Observability & compliance built-in

+ Apigee for API management | Vertex AI for model serving | Cloud IAM for security

ADK Development Workflow

From local development to production in three steps

1

Develop Locally

CLI + Web UI for rapid agent prototyping and debugging

→

2

Evaluate & Test

Built-in evaluation framework with debugging tools

→

3

Deploy to GCP

One-command deploy to Cloud Run or GKE with containerization

Why ADK Matters:

- Unified enterprise AI framework
- All GCP services no duplication
- Enterprise app connectivity
- Production-ready from day one

Why ADK for SRE?

Traditional Alerting

- Static threshold rules
- Manual investigation per alert
- Context switching between tools
- No memory of past incidents
- Linear, single-threaded process

ADK-Powered Agent

- AI-driven pattern recognition
- Parallel multi-agent investigation
- Unified tool access via MCP servers
- TF-IDF incident memory & matching
- Orchestrated, concurrent workflows

Incident Response Expectation

1. Get alert
2. Check dashboard
3. Find root cause
4. Fix it
5. Go back to sleep

Incident Response Reality

1. Get alert
2. Which cluster was this again?
3. VPN dropped
4. Wrong terminal, that was prod
5. Slack: "is anyone else seeing this?"
6. It was DNS. It's always DNS.
7. Sun is rising

Architecture

Multi-agent design with ADK orchestration & MCP tools

Plan A — The Dream

Original Vision

Full multi-agent MCP architecture — 1 orchestrator + 5 sub-agents + 5 MCP servers

ADK Root Orchestrator + 5 Sub-Agents

Powered by Gemini

↓ All tools via MCP ↓

GC Observability MCP	Google GKE MCP	Grafana MCP	Slack MCP	Notifications MCP
Cloud Logging Monitoring, Trace	Pod status events, deploys	Alerts, dashboards annotations	Channel alerts thread updates	Email, Telegram WhatsApp
official	official	official	custom	custom

Reality Check: **GKE & Observability MCPs are in preview/beta. In testing:**

MCP connections hanging/timing out • ADK TaskGroup/ExceptionGroup errors on MCP failures
Investigations stuck at "Triage" phase • httpx.AsyncClient connection pool issues

Plan B — Simple Mode

Working Today

1 agent + 10 kubectl FunctionTools — no MCP, no sub-agents, just kubectl

INGESTION

Grafana / Alertmanager / PagerDuty → Webhook Receiver → Dedup

▼

ADK AGENT *Single agent — no orchestrator, no sub-agents*

Single Agent (Gemini 3.1 Pro Preview)

▼ **kubectl subprocess calls** ▼

10 kubectl FunctionTools *Read-only RCA — no write operations*

get_pods	get_events	get_deployments	top_pods	get_pod_logs
describe_pod	get_deployment_history	get_namespaces	get_nodes	get_resource_yaml

▼

Output: Root Cause Analysis (read-only)

Root Orchestrator

The brain of the operation — powered by Gemini

Alert Received | **Root Orchestrator** | **Parallel Sub-agents** | **Synthesize & Act**

How it works:

- Receives deduplicated alert with fingerprint and context
- Delegates to Log Analyst, Deployment Correlator, Runbook Retriever in parallel
- Synthesizes findings using Gemini's reasoning to generate an RCA
- Proposes remediation actions, routed through the Policy Engine
- Drafts and sends notifications via Communication Drafter

Specialized Sub-Agents

Log Analyst

Queries GC Observability MCP for error patterns, stack traces, and anomalies

Deployment Correlator

Uses GKE MCP to check pod health, events, deployments, and what changed

Runbook Retriever

Searches Grafana MCP for alerts, dashboards, and past incident runbooks

Communication Drafter

Generates RCAs and sends via Slack MCP + Notifications MCP

MCP Servers: The Tool Layer

Model Context Protocol — how agents talk to infrastructure

Official MCP Servers

Google Cloud Observability MCP

Cloud Logging, Monitoring, Trace — remote, IAM auth

Google GKE MCP

Pod status, events, deployments — remote

Grafana MCP

Alerts, dashboards, annotations — stdio, token

Custom MCP Servers

Slack MCP

Post incident alerts, thread updates, channel notifications — custom built

Notifications MCP

Email (SMTP), Telegram, WhatsApp multi-channel delivery — custom built

"How many agents does your system have?"

Yes.

Log Analyst

Deployment Correlator

Runbook Retriever

Communication Drafter

Root Orchestrator

Anomaly Detector

Intelligence Layer

Incident Memory

- TF-IDF vectorization of past incidents
- Cosine similarity matching
- Learns from every investigation
- Surfaces relevant runbooks automatically

Policy Engine

4 approval levels:

- Auto-approve — safe actions (restart pod)
- Env-gated — auto in dev, manual in prod
- Always-approve — human required
- Blocked — never auto-execute

Ingestion Pipeline

Grafana | Alertmanager | PagerDuty | Generic

↓

Webhook Receiver

FastAPI on :8090
normalizes payload format

Fingerprint & Dedup

compute_fingerprint()
sliding window dedup

K8s Event Watcher

Continuous stream of
cluster events as context

When official MCP servers exist for GKE and Observability:

"npm install and done!"

When you need Slack + Notifications MCP:

"It ain't much, but it's honest work"

(400 lines of Python later...)

Before MCP:

"Let me write 47 custom API integrations"

After MCP:

"npm install @google-cloud/gcloud-mcp"

Official Google Cloud + Grafana MCPs = plug and play

A moment of silence...

for the Demo Gods

"Please work. Please work. Please work."

It worked on my machine. Yesterday. Once.

Live Demo

Let's break things and watch the agents investigate

Demo Scenarios

1	**memory-leak**	Gradual memory increase until OOM
2	**cpu-spike**	Sudden CPU utilization spike
3	**cascading-failure**	Multi-service dependency chain failure
4	**config-drift**	Configuration mismatch detection
5	**oom-kill**	Container killed by kernel OOM

Demo Commands

Quick reference for the live demo

Start agent	`$ nightops agent watch`
Interactive	`$ nightops agent run --interactive`
Trigger	`$ ./scripts/demo-gke.sh --scenario cpu-spike`
Dashboard	`$ nightops dashboard`
Verify	`$ nightops verify`

What to Watch For

During the live demo, pay attention to these key moments

1	**Alert arrives via webhook**	Watch the dedup engine filter noise
2	**Orchestrator delegates**	Sub-agents spin up in parallel
3	**MCP servers query infra**	GKE, Observability & Grafana MCPs fetch live data
4	**RCA generation**	Gemini synthesizes findings into root cause
5	**Notification sent**	Slack MCP + Notifications MCP deliver the report

Production:

"This is fine."

TheNightOps:

"Already identified root cause, drafted RCA, and notified the team. Go back to sleep."

MTTR with manual investigation:

45 minutes

MTTR with TheNightOps:

< 3 minutes

...and your SRE actually got to sleep

Key Takeaways

ADK makes multi-agent systems accessible

Clean abstractions for orchestration, sub-agents, and tool integration

MCP is the universal tool protocol

Official Google Cloud & Grafana MCPs + custom servers = plug and play

Gemini excels at reasoning over SRE data

Pattern recognition across logs, events, and deployment history

Start with guardrails, not YOLO

Policy Engine ensures safe, graduated automation from day one

The real takeaway from this talk:

Let the agents handle 3 AM.

You deserve sleep.

(Your on-call rotation will thank you)

Get Started

1	**Clone & Install**	`git clone github.com/nomadicmehul/TheNightOps`
2	**Configure**	`cp config/.env.example config/.env`
3	**Verify**	`nightops verify`
4	**Run Demo**	`./scripts/demo-gke.sh --scenario memory-leak`
5	**Explore**	`nightops agent run --interactive`

Thank You!

Questions & Discussion

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github.com/nomadicmehul/TheNightOps

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